UVA DS2003 Communicating with Data
Fall 2026

Storytelling with Data
Narrative Visualization

Sarah Groves
based on materials by Hannah Bako and a lot of other people

Where failed data stories go wrong

Summary from last class

- Explanatory visualization is about communicating a story that you found in the data
  - It is not about manipulating the data to tell a story that you already think is true!
  - Effective storytelling should guide the viewer to the core message and allow for a restrained amount of exploration
  - In Western cultures, we read left to right and top to bottom, so most important information should be top left
- Use clear annotations and legends to highlight the main story
- Too much data/chronological “as you made them” charts will not be effective

Think | Pair | Share

- Think of a time someone tried to tell you or others a story with data, and it failed.
- What do you think went wrong?
- Why was the intended message not communicated effectively?

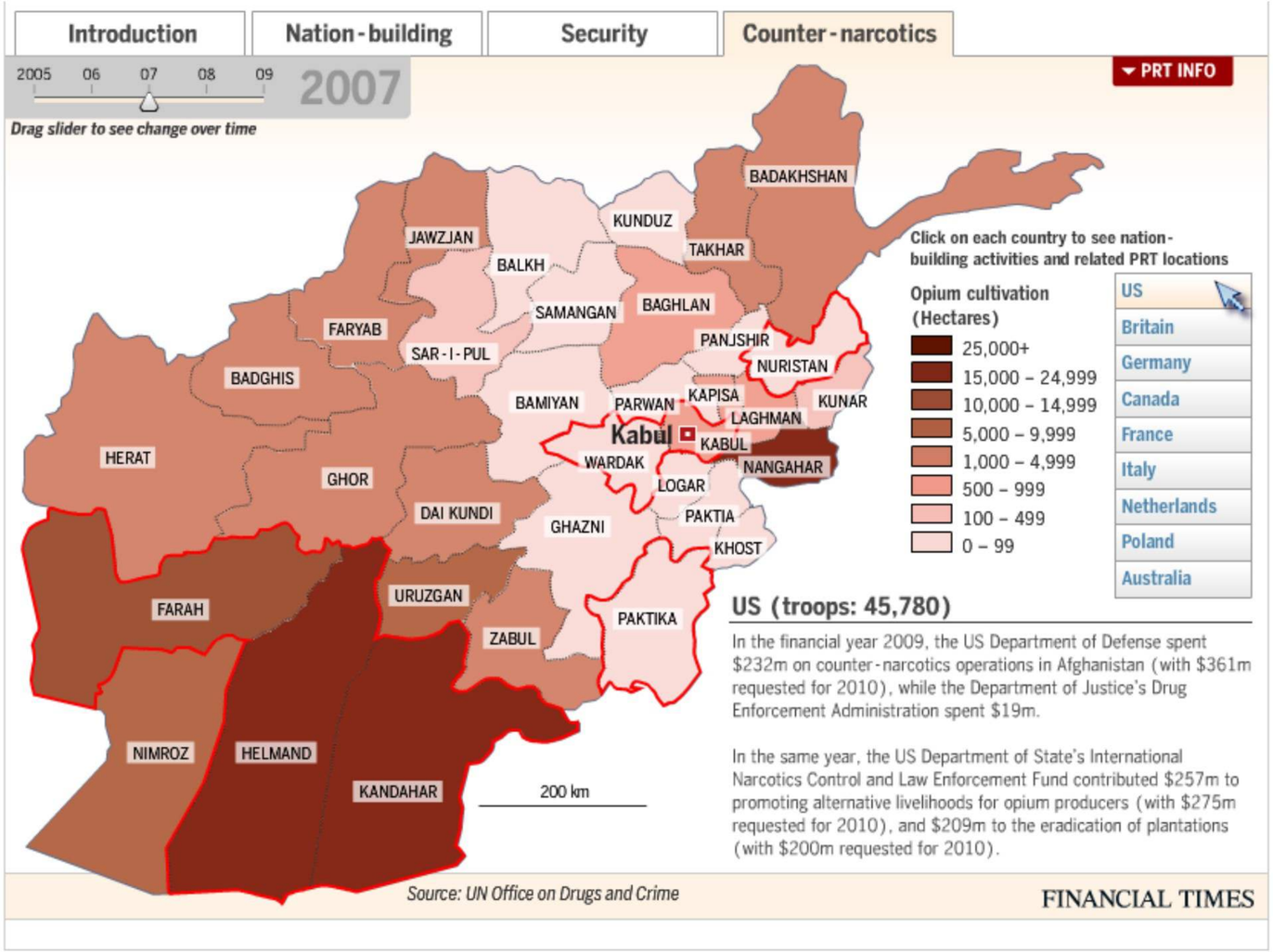
Afghanistan: behind the front line

Intended purpose of the graphic: to establish indicators of success by which to evaluate the development work being done in Afghanistan

- Semantically consistent color

- Details on demand

- However, it does not guide the viewer through the data at all → too much flexibility → viewer can't reach meaningful conclusions intended by the designers



The graphic "suffers by putting exploratory power into the hands of the viewer without sufficient guidance"

Minnesotan Employment Explorer

Context: in July 2008, Minnesota was in recession, so Minnesota Public Radio (MPR) released “Minnesota Slowdown:” stories detailing the state’s economic malaise and shares interviews with residents hit hard by the downturn.

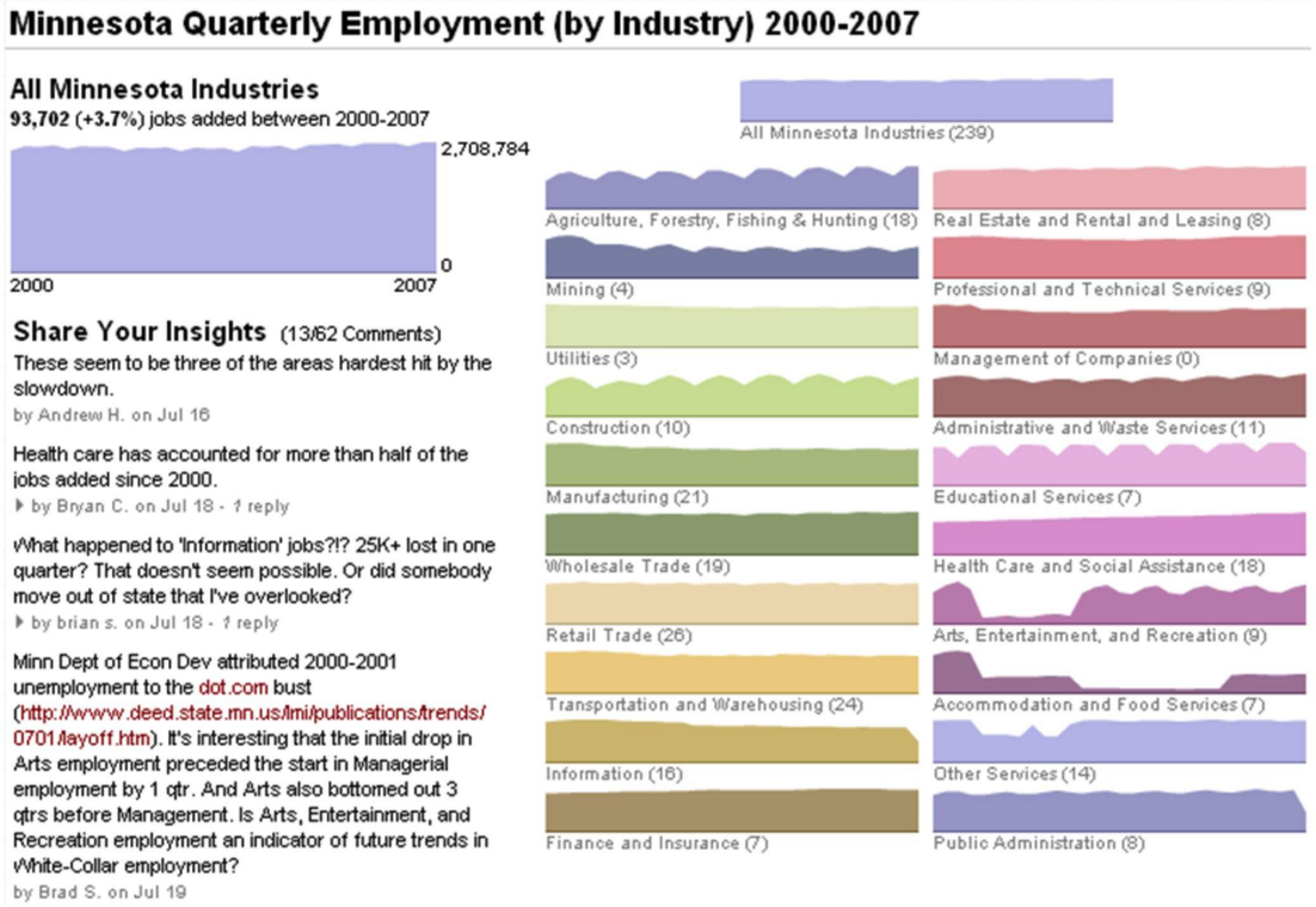
Intended purpose: engage readers in finding and telling their own stories in the data.

Only 23 people engaged, and not Minnesotans

- No tutorial and too much data → confused users

- No rich, emergent story

- Disconnected from the narrative (graphic showed 2000-2007)



Narrative Tactics

A design space for narrative visualization

- Segel & Heer describe every narrative visualization through three sets of choices:
  - Genre: the form the story takes (seven genres)
  - Visual narrative tactics: visual devices that guide the viewer's attention
  - Narrative structure tactics: non-visual mechanisms that shape the path, the interaction, and the message

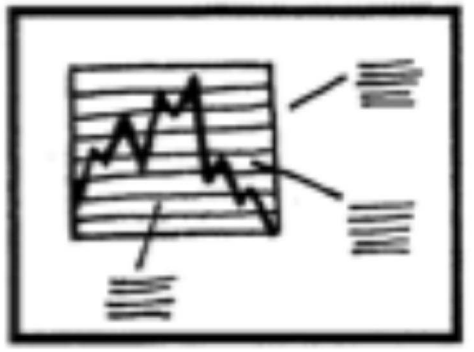
Segel, E., & Heer, J. (2010). Narrative visualization: Telling stories with data. IEEE Transactions on Visualization and Computer Graphics, 16(6), 1139–1148.

Types of Data Stories

Seven Genres



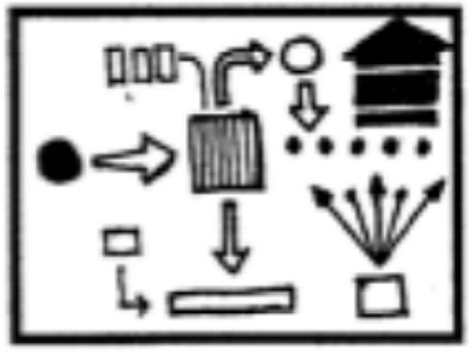
Magazine Style



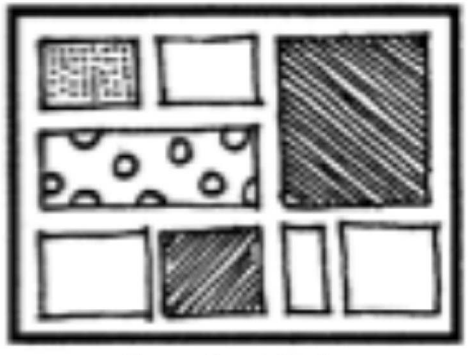
Annotated Chart



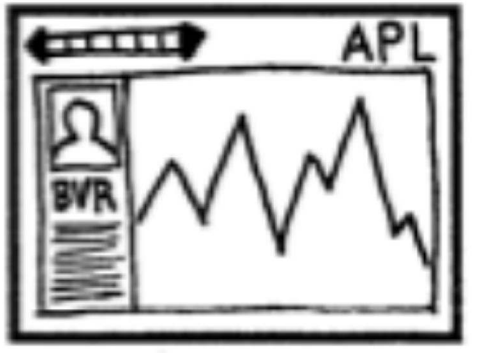
Partitioned Poster



Flow Chart



Comic Strip



Slide Show



Film/Video/Animation

Magazine Style



Business

Also in this section
68 Sailing companies
69 The recovery of General Motors
69 Business crime in China
70 Italian retail
70 Japan's sake firms on the move
71 Brazilian tobacco
71 Schumpeter: Overstretched

Privacy and the internet

Lives of others

Facebook and Google face a backlash from users and regulators alike, over the way they have handled sensitive data

JENNIFER STODDART, Canada's privacy commissioner, is not on Facebook. In August 2009 the social networking site struck a deal, agreeing to change its policies within a year to comply with the country's privacy law. It now says Ms Stoddart, the only one appears to be dragging an unimportant part of the deal, which involved giving users a clear and easy-to-understand explanation of the tool it employs to pass data with third parties. "It doesn't seem to me that foot-dragging is the right direction on this issue," she says. Having had, without a change of course, the firm could soon become the subject of another formal investigation by her organisation.

Facebook is not the only internet giant to provoke the ire of data watchdogs. Google endured withering criticism this week following news that it had recorded some personal communications sent over unencrypted Wi-Fi data networks in homes and offices in some 30 countries. On May 11th Peter Schaar, Germany's federal commissioner for data protection, called for an independent investigation into Google's behaviour, claiming that it had "deeply disrespected essential rules in the development and usage of software."

The cases highlight rising tension between guardians of privacy and internet firms. And they reflect concern among web users about how private data can be made public. Several prominent internet types, such as Cory Doctorow, a science-fiction author, and Ian Lopatin, a producer, have abandoned Facebook. Sites such as QuitFacebookDay.com are urging others to do so next month (May 31st is the unofficial Facebook-quitting day).

There is likely to be a big exodus in store for Facebook, which is poised to claim half a billion members and which draws even more visitors as a whole to its site (see chart). Its owners have been ranked at the company's headquarters in Silicon Valley, where bosses are reading over how to respond. Several senior folk are now hinting that Facebook will soon roll out simpler privacy controls to make it easier to keep more data hidden. MySpace, a rival, is already making it controls simpler in an effort to woo disaffected Facebookers to its service.

A revolt over Facebook's handling of privacy has been brewing for some time. In December the social network changed the default settings on its privacy controls so that individuals' personal information would be shared with "everyone" rather than "selected friends". Facebook argued this reflected a shift in society towards greater openness and noted that users could still adjust privacy settings back again. But incensed privacy activists lobbied for it to be reversed.

The switch should not have come as a surprise. Early on, many social networks

For daily analysis and forums on business, visit economist.com/business-finance

expose flaky to ugly privacy policies in order to attract and motivate users. But as more join, controls are gradually loosened to encourage more sharing. As people share more, Facebook can increase the traffic against which it sells advertising. And the more it learns about users' likes and dislikes, the better it can target ads that generate hundreds of millions of dollars.

Problems grew louder still following a developers' conference last month at which Mark Zuckerberg, Facebook's boss, announced yet another series of policy changes. One that caused confusion was an "instant personalisation" feature that lets certain third-party websites access Facebook data about people visit. Critics say that Facebook has made it risky to disable this feature, which may explain why so Stoddart dislikes it so much.

European officials are grumbling about Facebook, too. This month a group of data-protection experts who advise the European Commission wrote to the social network, calling its decision to loosen the default settings "unacceptable". And as the in

Annotated Chart

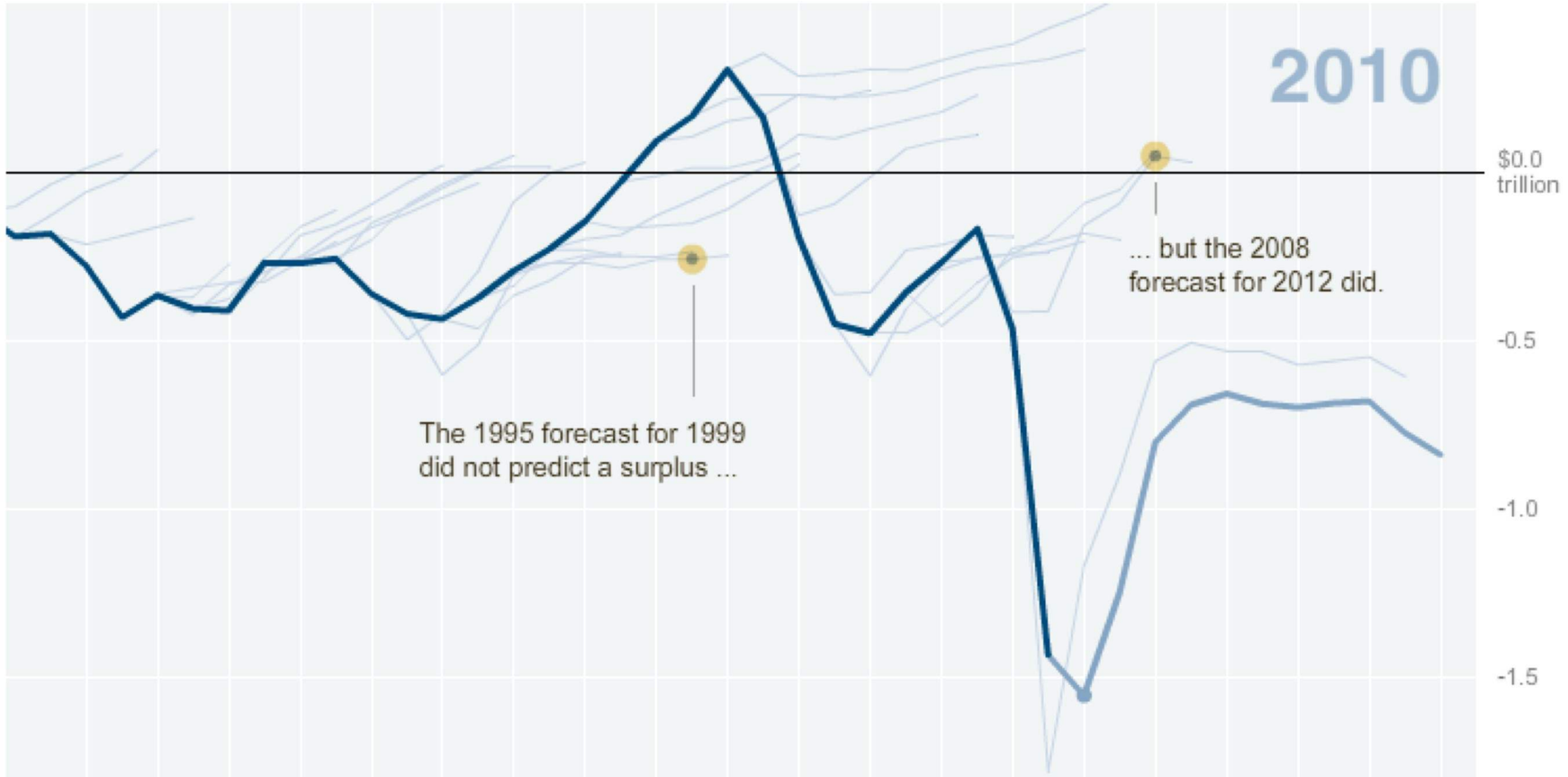
Published: February 2, 2010

Budget Forecasts, Compared With Reality

Just two years ago, surpluses were predicted by 2012. How accurate have past White House budget forecasts been?

Latest forecast

Today, with a better understanding of the severity of the economic downturn, the deficit situation is much more dire.



Annotated Chart + Slideshow

Published: February 2, 2010

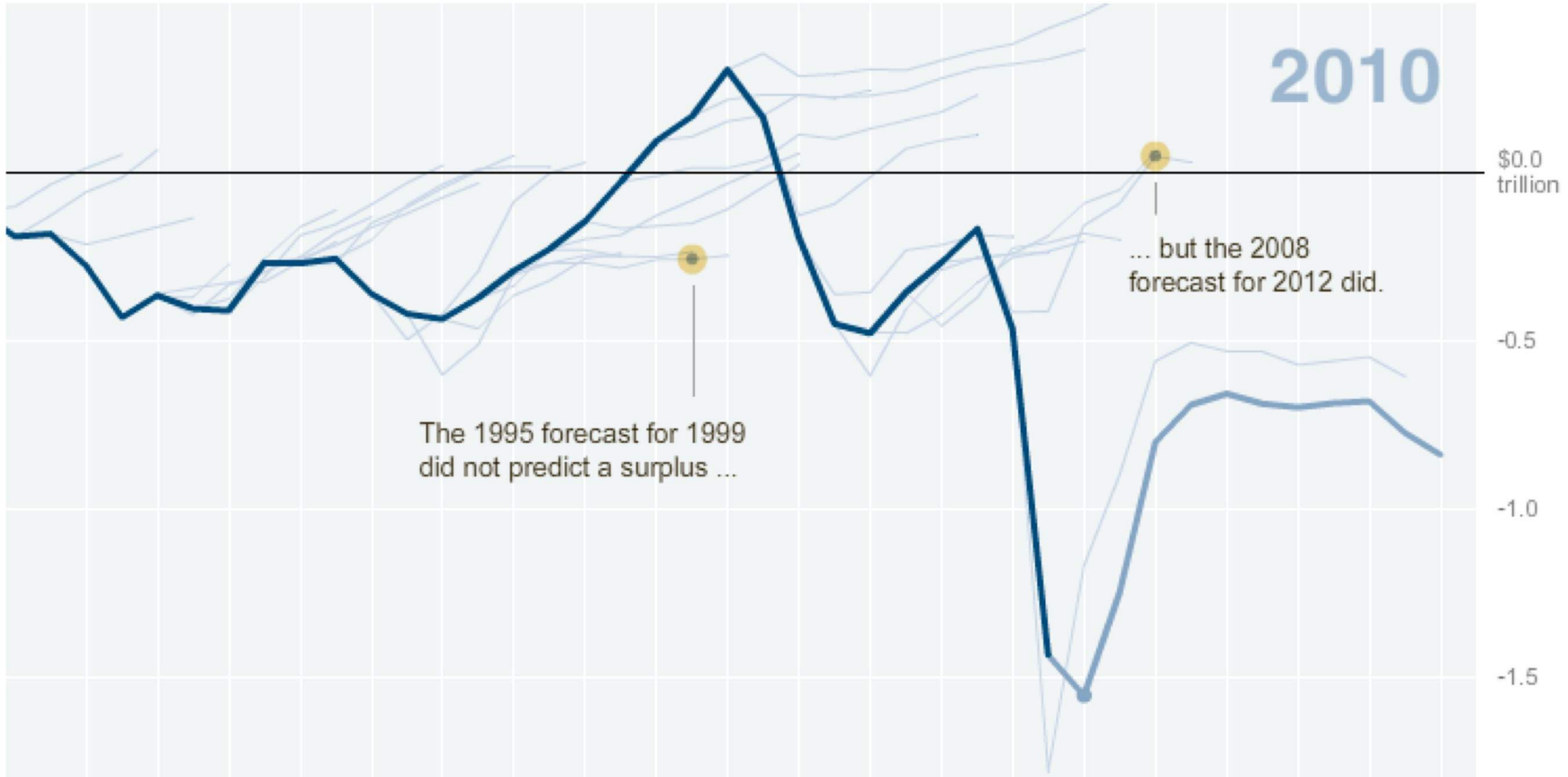
Budget Forecasts, Compared With Reality

Just two years ago, surpluses were predicted by 2012. How accurate have past White House budget forecasts been?

1 2 3 4 5 6 NEXT ▸

Latest forecast

Today, with a better understanding of the severity of the economic downturn, the deficit situation is much more dire.

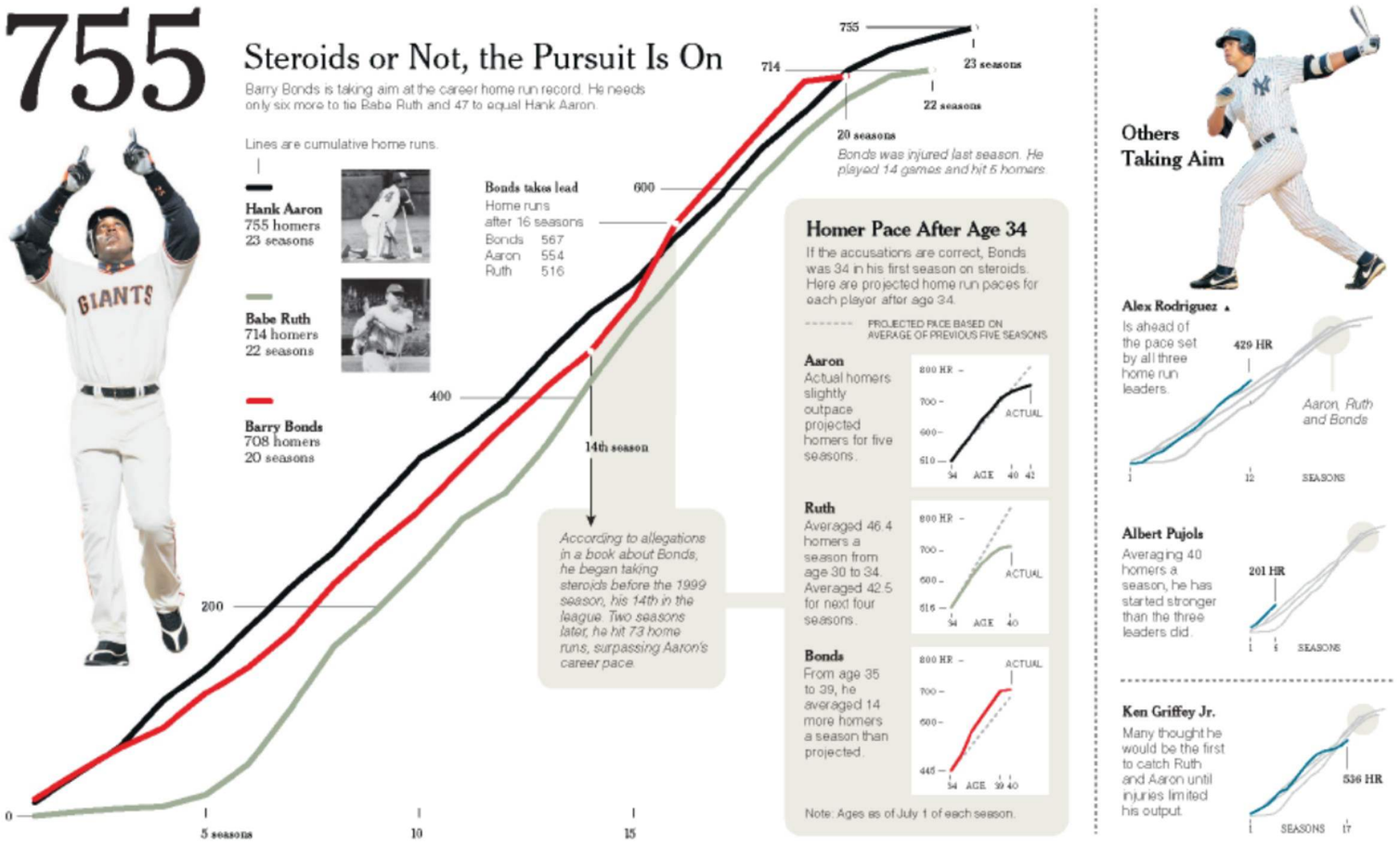


Annotation Types

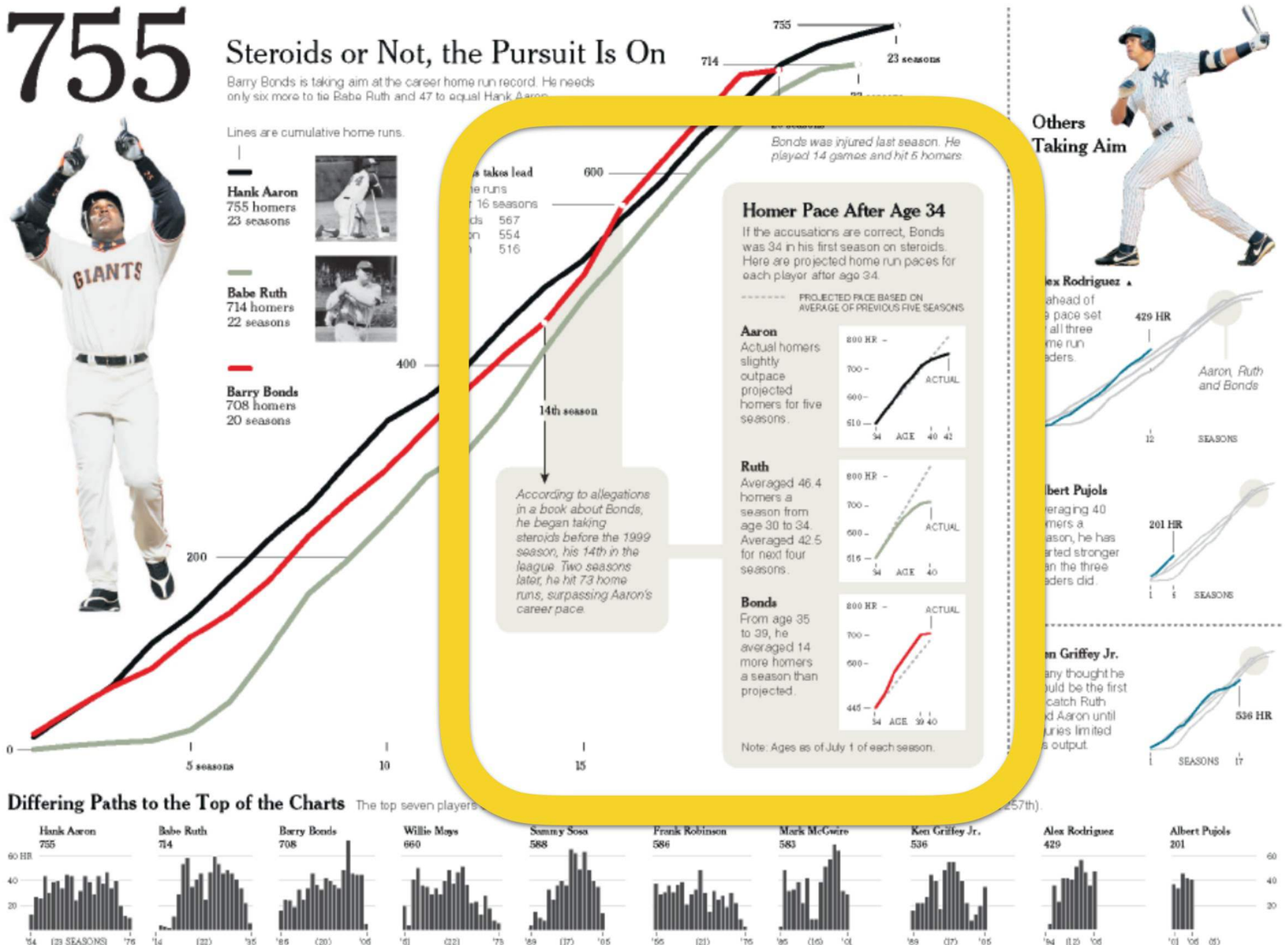


enclosure connector text glyph color indicator geometric

Partitioned Poster

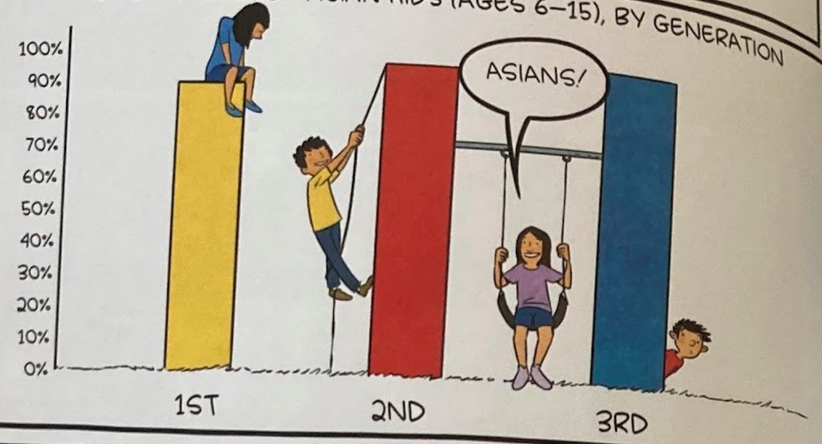


Partitioned Poster + Flow Chart

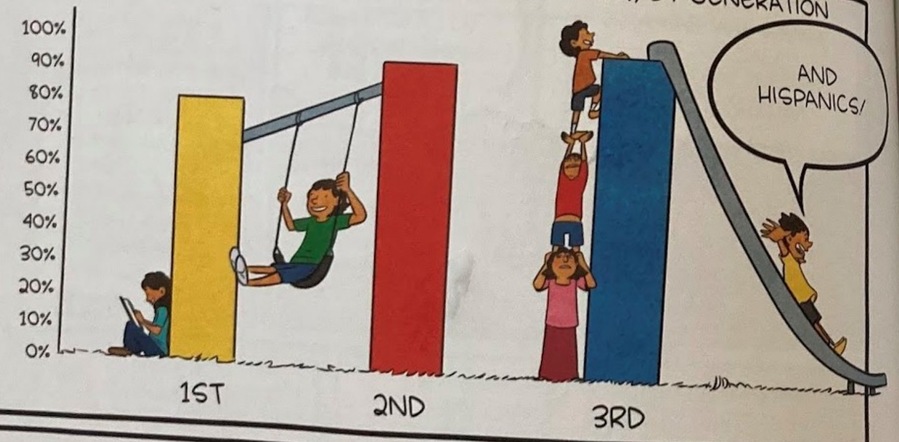


A CENTURY LATER, LITTLE HAS CHANGED. A LARGE MAJORITY OF ***FIRST***-GENERATION KIDS SPEAK ENGLISH WELL. RATES FOR LATER GENERATIONS APPROACH 100%. THIS IS TRUE EVEN FOR GROUPS THAT ALLEGEDLY "DON'T LEARN ENGLISH," LIKE...

ENGLISH SKILLS—ASIAN KIDS (AGES 6–15), BY GENERATION



ENGLISH SKILLS—HISPANIC KIDS (AGES 6–15), BY GENERATION

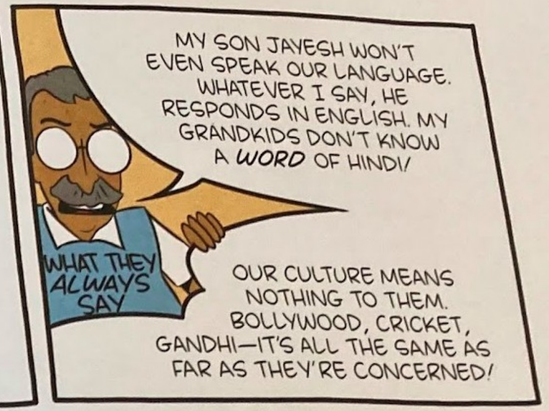


NOTE: SINCE WE'RE TALKING ABOUT KIDS, THESE NUMBERS ***UNDER***STATE ENGLISH FLUENCY. MY WIFE, WHO IMMIGRATED WHEN SHE WAS SEVEN, NOW SPEAKS ROMANIAN WITH AN AMERICAN ACCENT, NOT ENGLISH WITH A ROMANIAN ACCENT.

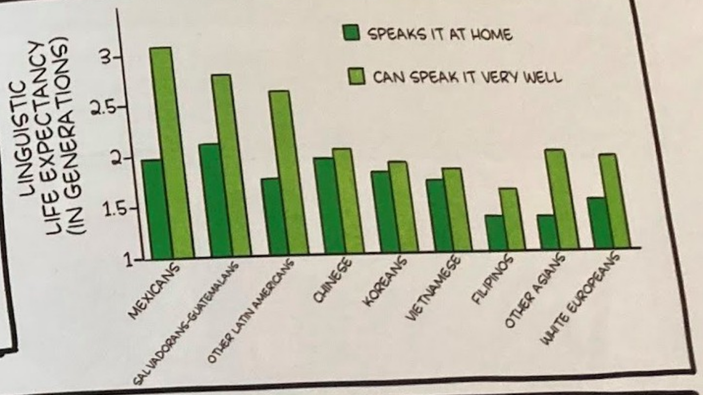


IF YOU WANT THE ***REAL*** STORY OF ASSIMILATION, ASK AN IMMIGRANT PARENT...

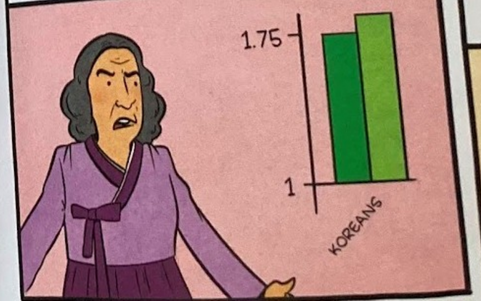




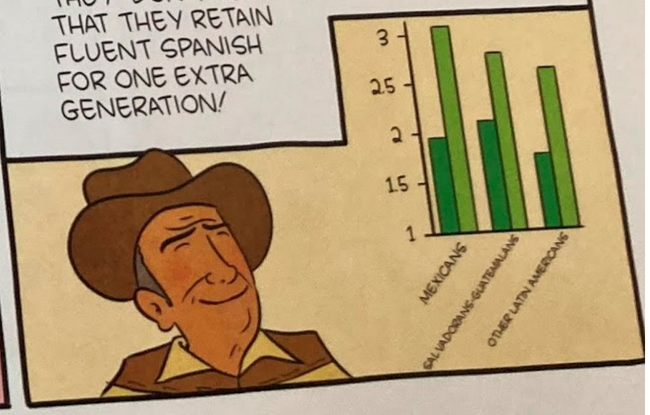
RESEARCHERS ACTUALLY MEASURE "LINGUISTIC LIFE EXPECTANCIES"—THE AVERAGE NUMBER OF GENERATIONS A FOREIGN LANGUAGE SURVIVES. HERE ARE THE NUMBERS FOR "MULTICULTURAL" SOUTHERN CALIFORNIA:



ON AVERAGE, FOR EXAMPLE, IT TAKES LESS THAN TWO GENERATIONS FOR KOREAN IMMIGRANTS TO LOSE KOREAN FLUENCY.



WHAT'S SPECIAL ABOUT HISPANICS IN PLACES LIKE CALIFORNIA ISN'T THAT THEY "DON'T LEARN ENGLISH," BUT THAT THEY RETAIN FLUENT SPANISH FOR ONE EXTRA GENERATION!



Visual narrative tactics

- Visual structuring*: where am I in the story?*
  - Orient the viewer early: establishing shot, checklist, consistent visual platform
  - Track progress: progress bar, timeline slider
- Highlighting*: where should I look?*
  - Make an element stand out from its surroundings: color, motion, framing, size, audio
  - Many are borrowed from film, art, and comics
- Transition guidance*: how did I get here?*
  - Move within or between scenes without disorienting the viewer: continuity editing, animated transitions, object continuity, camera motion

Narrative structure tactics
non-visual mechanisms

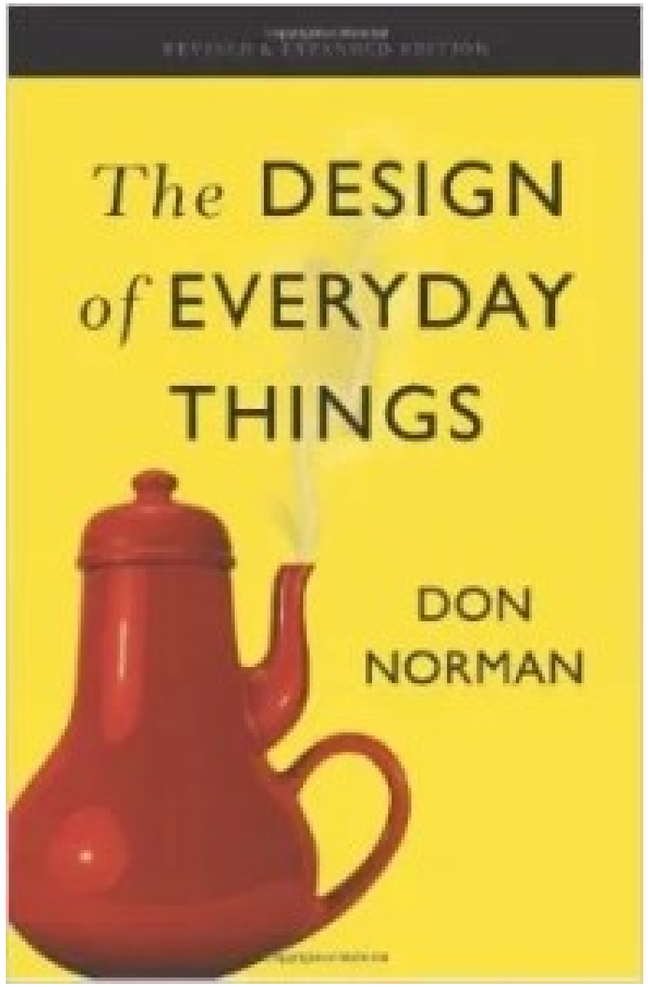
- Ordering*: what path do I take?*
  - *Linear* (author prescribes the path), *random access* (no path suggested), *user-directed* (reader picks among alternatives)
- Interactivity*: what can I do, and how do I learn it?*
  - Filtering, selecting, searching, navigating
  - Learned through explicit instruction, a tacit tutorial, or the initial configuration
- Messaging*: what am I being told?*
  - Short text: labels, captions, headlines, annotations
  - Longer text: articles, introductions, summaries

Project 2: your Nov 15 narrative outline should name your ordering, visual structuring, and messaging choices

Interactivity

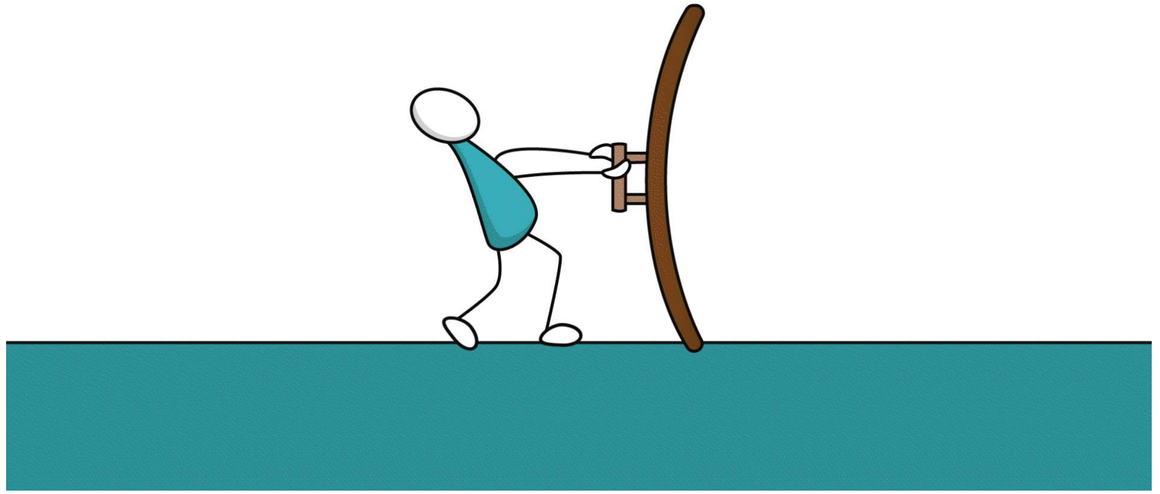
Affordance

- User’s perception of whether some action is possible



Affordance: Example





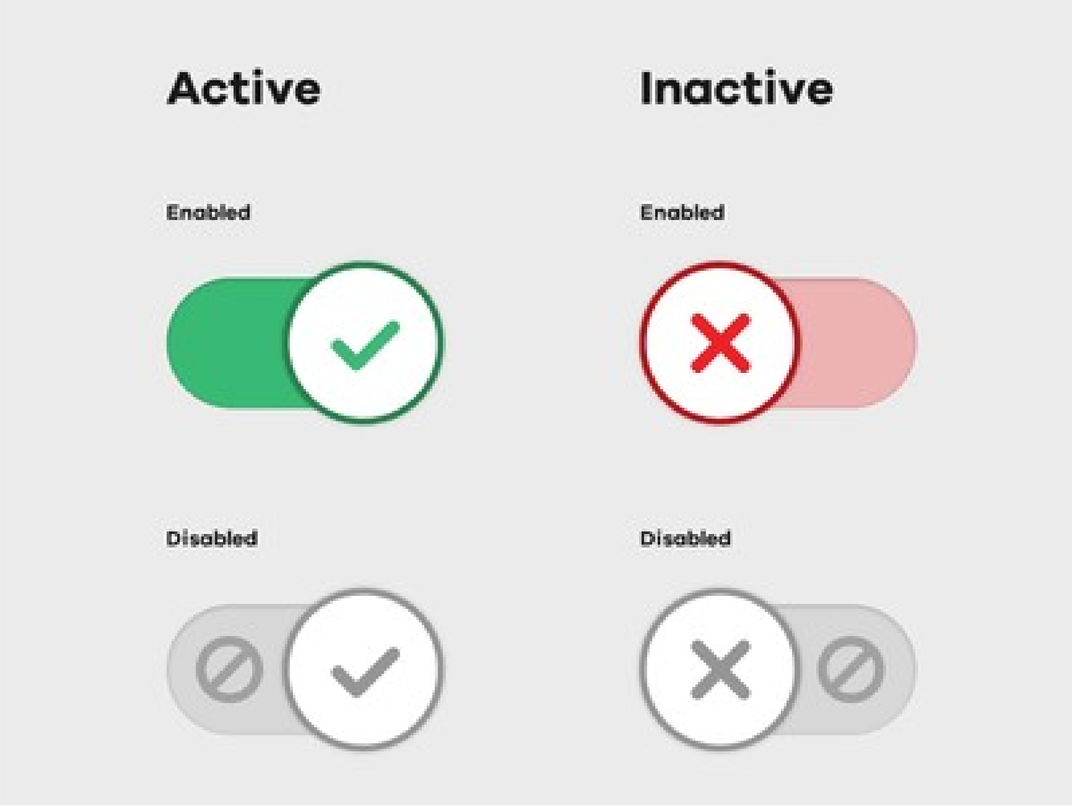
A Norman Door is one of those f*&king doors,
that has the same handle on both sides.
Giving you no indication as to if
...you should push it...
or pull it.

IMO, if a 'push' or 'pull' label is required, the design sucks.
└, The best design is intuitive.

- Elijah Szasz



Can you think of any examples in user interfaces?





Think | Pair | Share

- What's an example of a time *bad design* meant you couldn't figure out how to interact with something?
- Often, design is hidden and we only notice it when it fails

Interactivity in visualization

Information Visualization: Mantra

- Overview, zoom & filter, details-on-demand
- Overview, zoom & filter, details-on-demand
- Overview, zoom & filter, details-on-demand
- Overview, zoom & filter, details-on-demand
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- Overview, zoom & filter, details-on-demand
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- Overview, zoom & filter, details-on-demand
- Overview, zoom & filter, details-on-demand

Slide courtesy of Ben Shneiderman

Yi et al's Taxonomy

- Surveyed 59 papers and 51 systems
- Resulted in 311 individual interaction techniques implemented in Infovis systems
- Focus on user intent:
  what users achieve by using the interaction techniques
  rather than how the techniques provided by Infovis systems work
- Many different styles of interaction techniques serve a relatively small set of purposes

Seven Categories

- **Select**: mark something as interesting
- **Explore**: show me something else
- **Reconfigure**: show me a different arrangement
- **Encode**: show me a different representation
- **Abstract/Elaborate**: show me more or less detail
- **Filter**: show me something conditionally
- **Connect**: show me related items

Selection

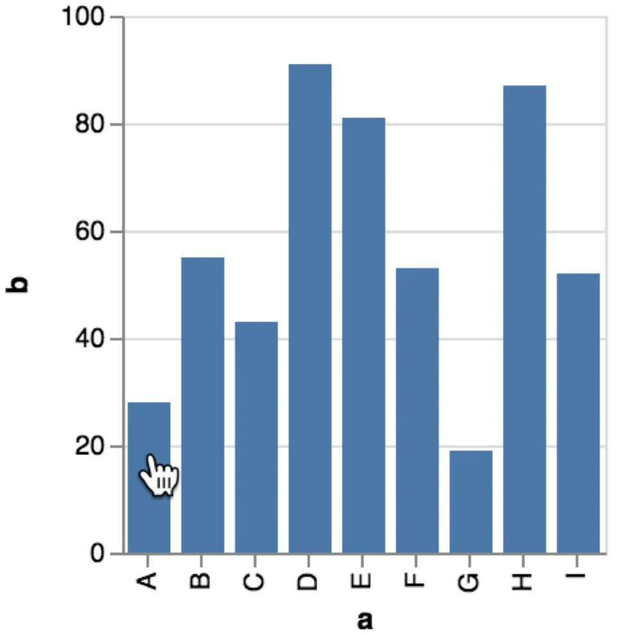
identify something as interesting

Select discrete data values.

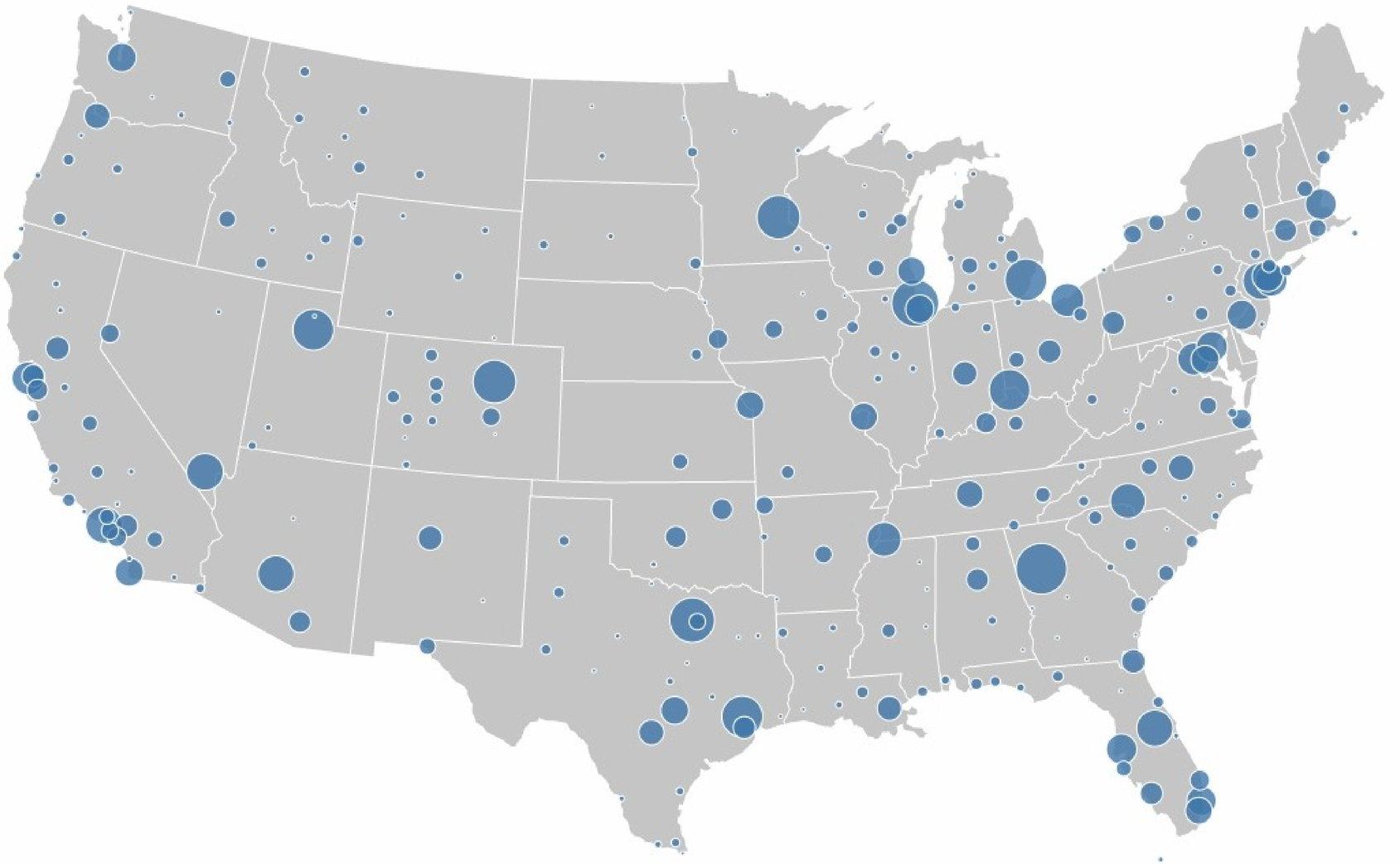
How many points are selected? 1, 2, 3, ...

How are points selected? Mouse hover, click, tap, ...

- Point Selection
- Mouse Hover / Click
- Touch / Tap
- Select Nearby Element (e.g., Voronoi-based Point Picker)



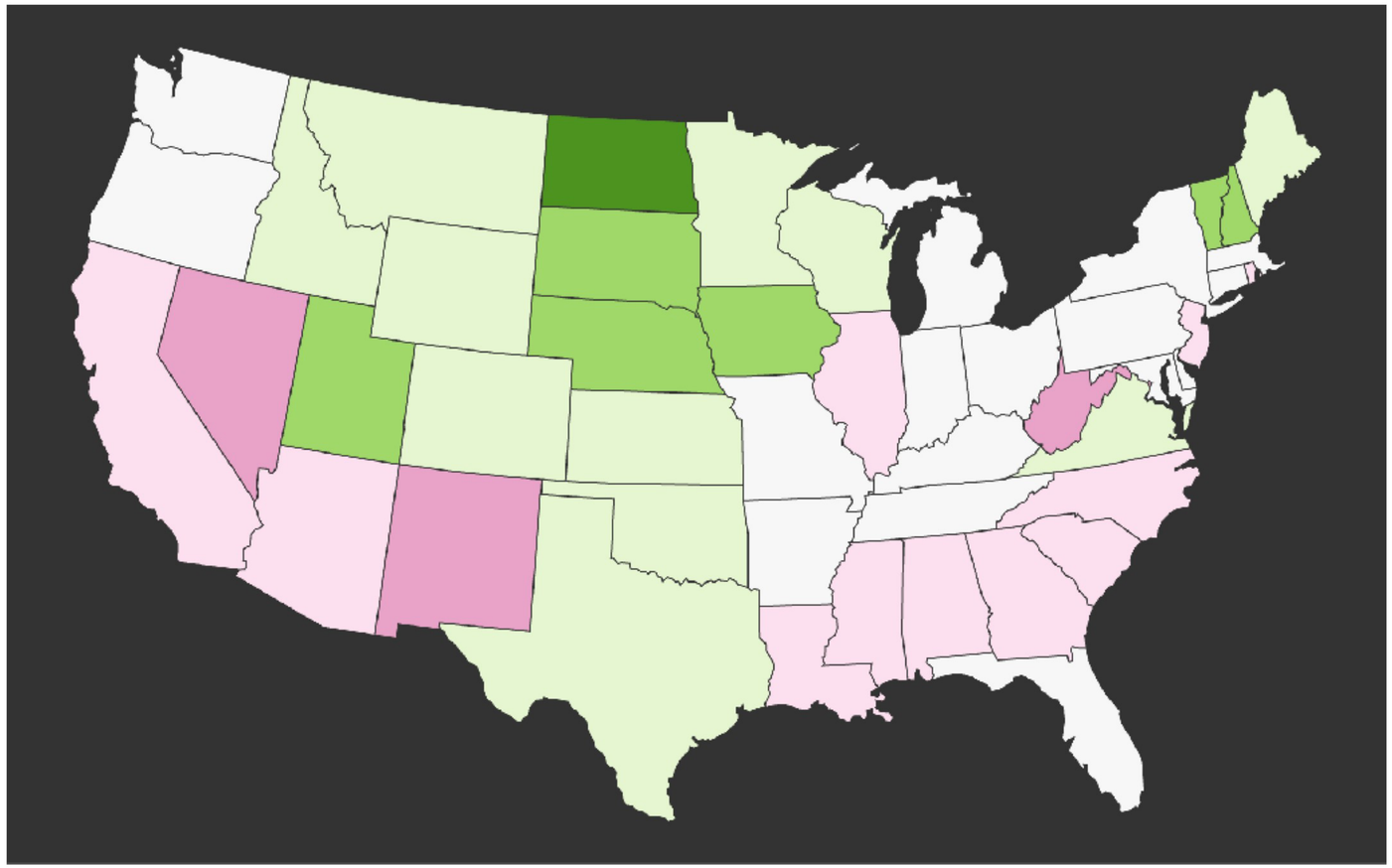
https://vega.github.io/vega-lite/examples/interactive_bar_select_highlight.html



http://mbostock.github.io/d3/talk/20111116/#14

Explore

show me something else



- Typically only viewing a subset of the data due to screen limitations
- Avoid complete changes in the data being viewed
- instead, some new data items enter the view as others are removed.

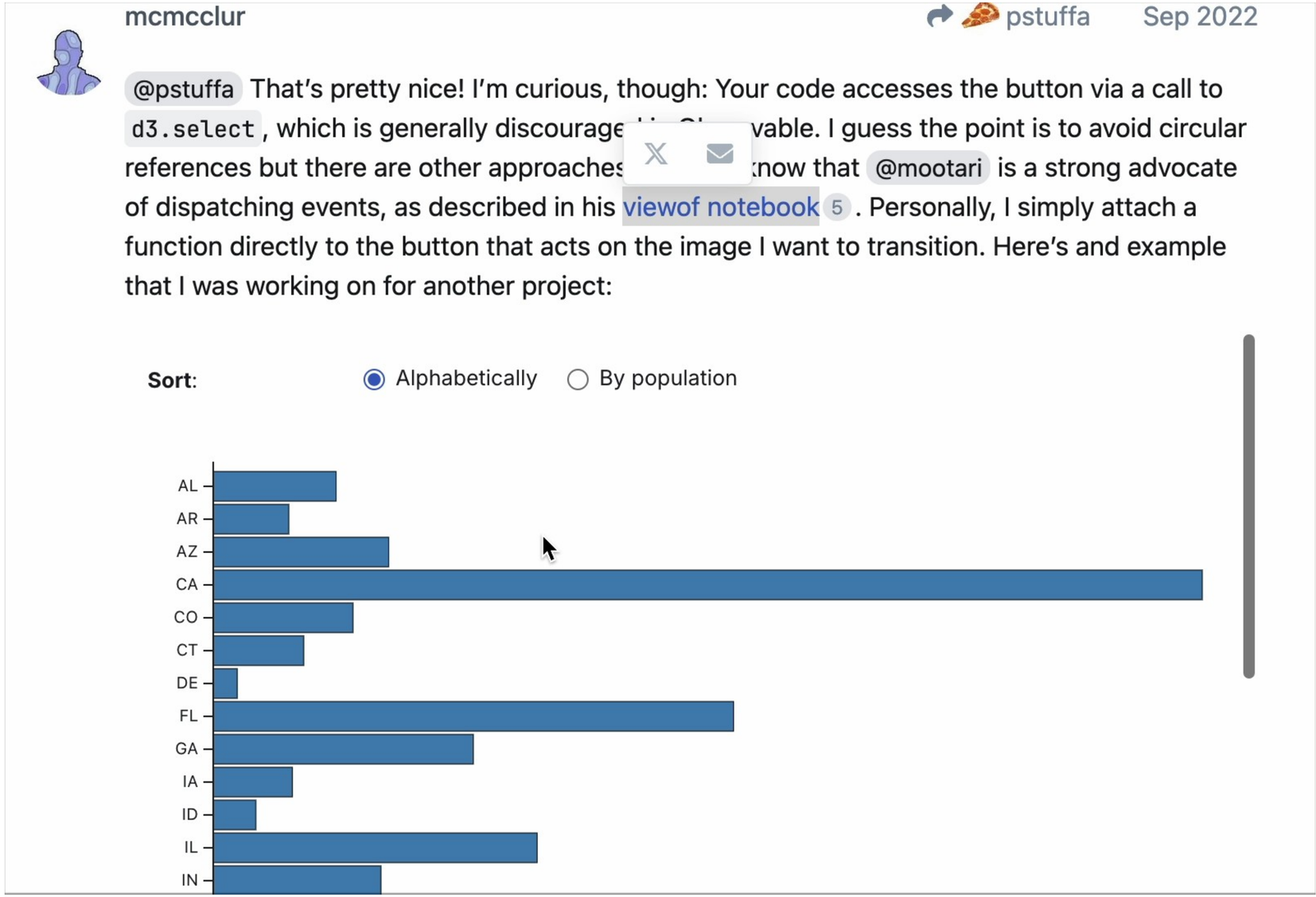
- Panning

https://bl.ocks.org/cmgiven/d39ec773c4f063a463137748097ff52f

Reconfigure

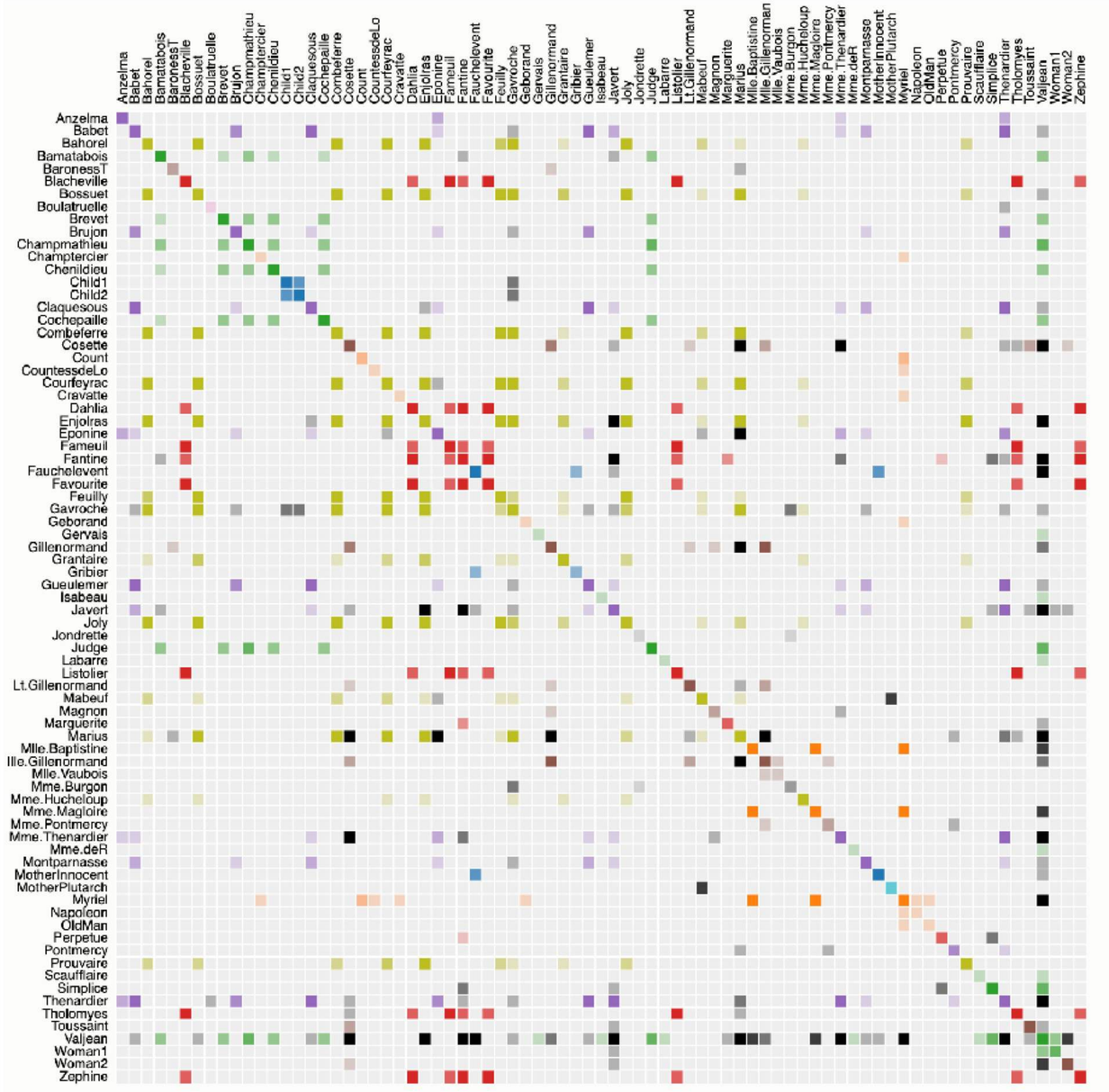
Sort Bars on click

- Changing the spatial arrangement of representations
- To reveal hidden characteristics of data and the relationships between them



Reconfigure

show me a different arrangement



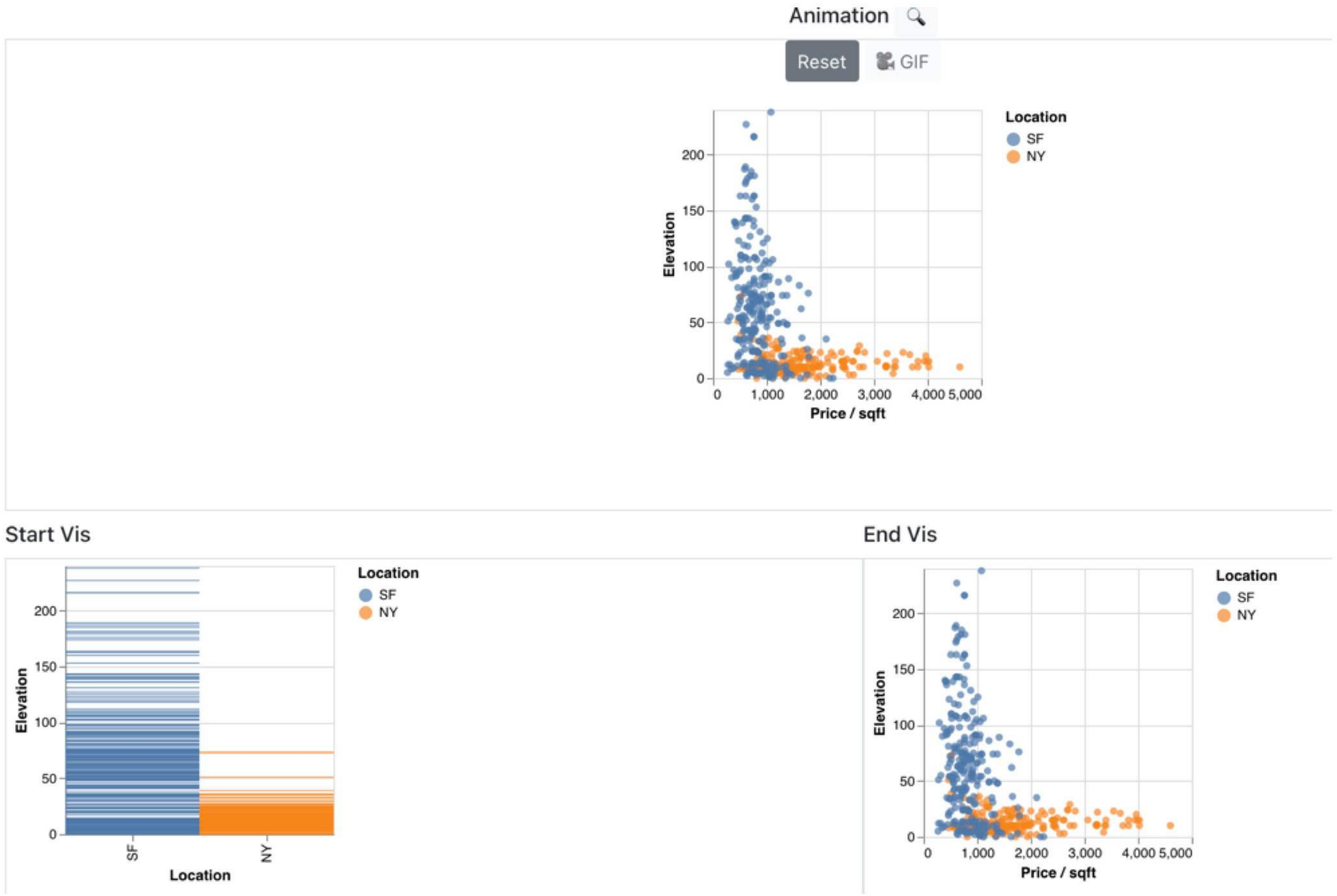
https://bost.ocks.org/mike/miserables/

Encode

show me a different visual representation

https://uwdata.github.io/gemini-editor/
-> dot plot to scatter plot

- Change the visual encodings
- Alter the fundamental visual representation of the data (e.g., color, size, and shape)



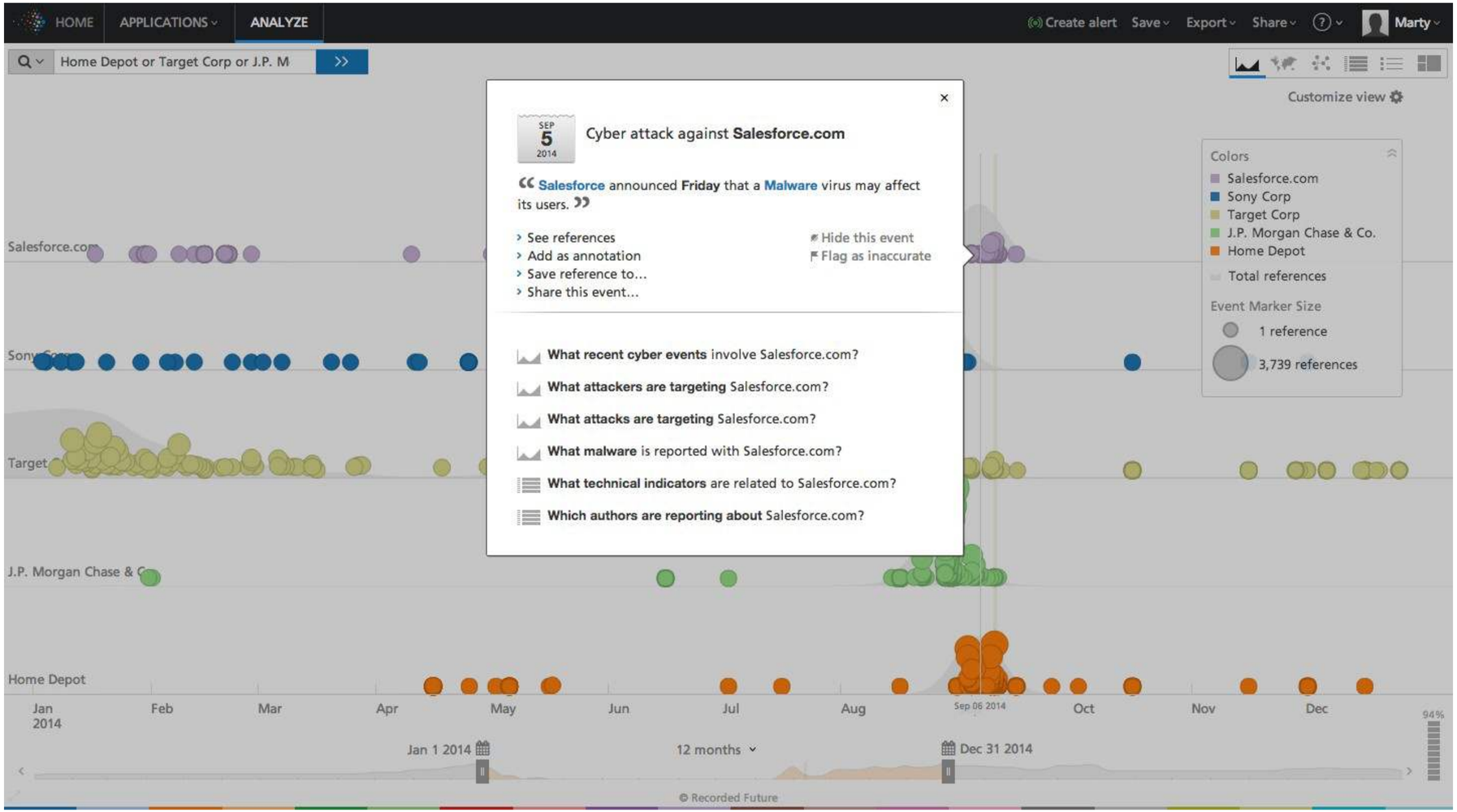
Abstract/Elaborate

Show more or less detail

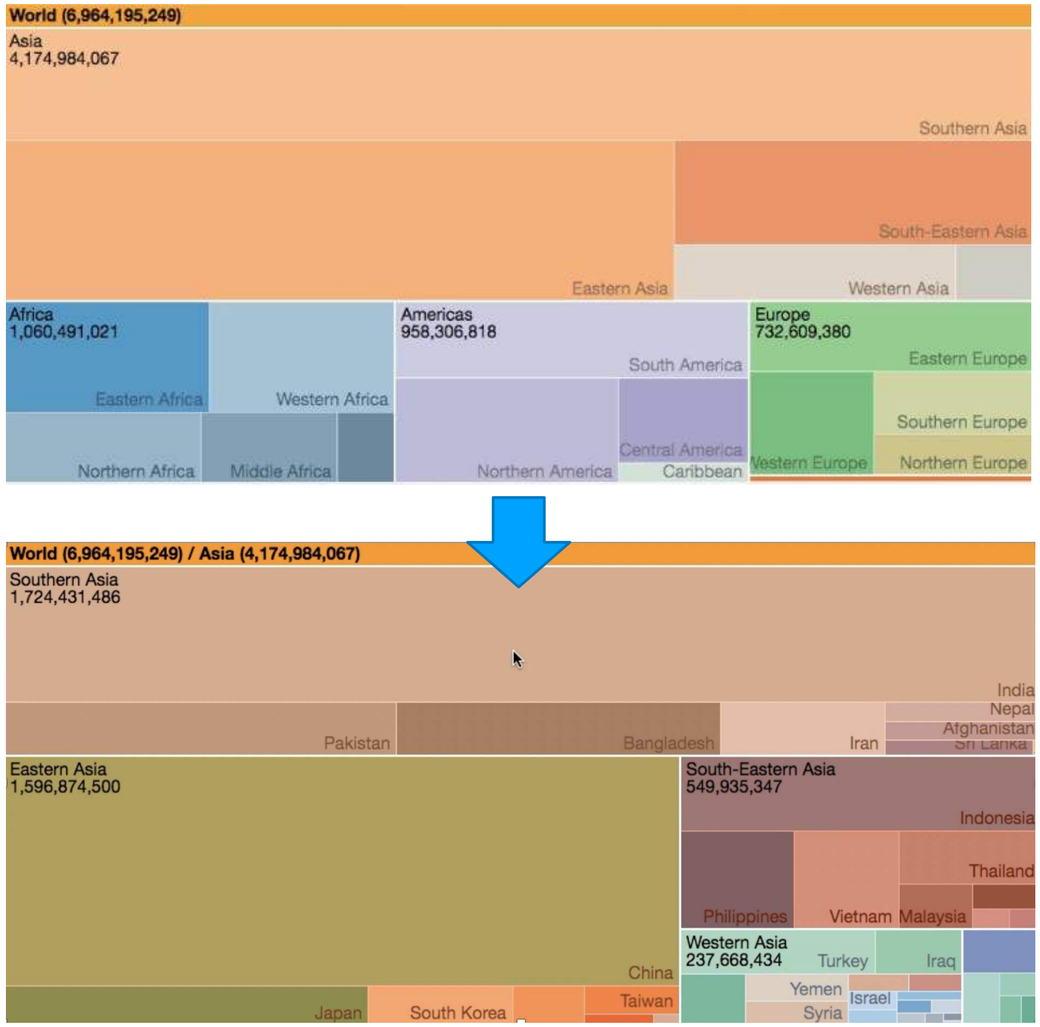


https://www.nytimes.com/interactive/2014/06/05/upshot/how-the-recession-reshaped-the-economy-in-255-charts.html

Detail-on-demand

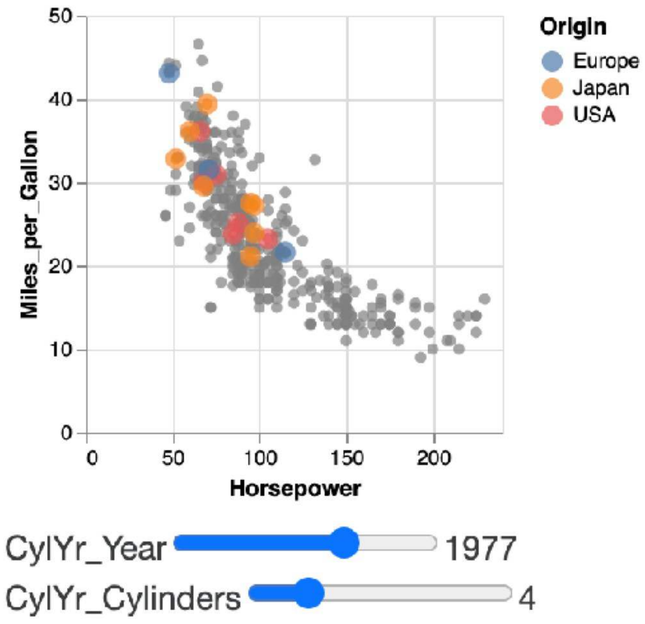


Drill down


World (6,964,195,249)
Asia
4,174,984,067
Southern Asia
South-Eastern Asia
Eastern Asia
Western Asia
Africa
1,060,491,021
Eastern Africa
Western Africa
Northern Africa
Middle Africa
Americas
958,306,818
South America
Central America
Northern America
Caribbean
Europe
732,609,380
Eastern Europe
Southern Europe
Western Europe
Northern Europe
World (6,964,195,249) / Asia (4,174,984,067)
Southern Asia
1,724,431,486
India
Nepal
Afghanistan
Pakistan
Bangladesh
Iran
Eastern Asia
1,596,874,500
China
Japan
South Korea
Taiwan
South-Eastern Asia
549,935,347
Indonesia
Thailand
Philippines
Vietnam
Malaysia
Western Asia
237,668,434
Turkey
Iraq
Yemen
Israel
Syria

Filter

show me something conditionally



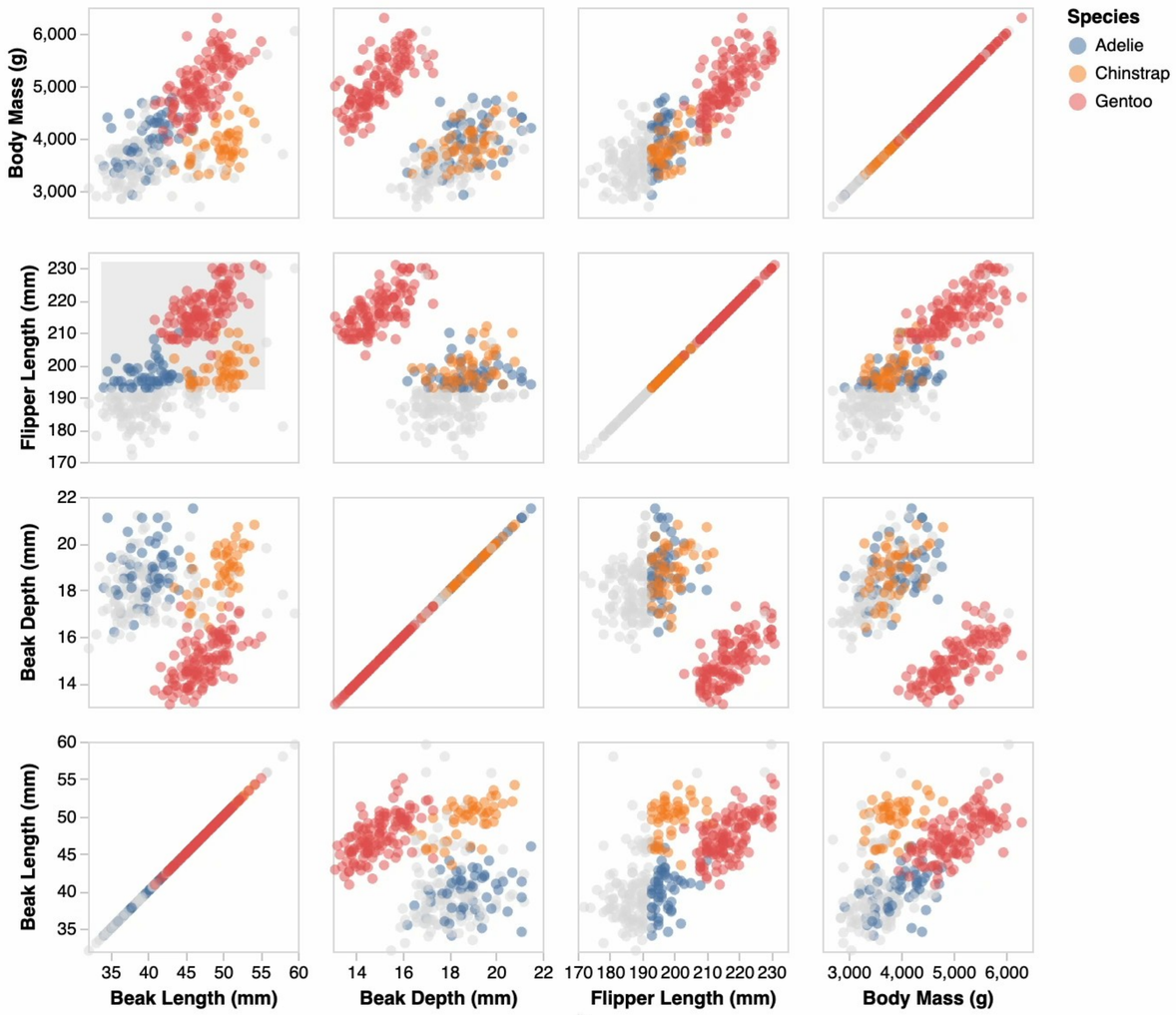
https://vega.github.io/vega-lite/examples/interactive_query_widgets.html

- change the set of data items being presented based on some specific conditions
- not changing perspective on the data
- actual data usually remain unchanged so that whenever users reset the criteria, the hidden or differently shown data items can be recovered

Connect

Show related items

- Brushing and Linking: selection is called a brush and related items are shown in linked views
- Link by tuple: explicitly identify and match the same data values across all views



View in Online Vega Editor

https://vega.github.io/vega/examples/brushing-scatter-plots/

Summary

- Narrative tactics: both visual and non-visual structure clarifies narrative visualization
- Types of data stories: for your Project 2, you will probably lean most on annotated charts, partitioned poster (infographic), flow chart, and slide show
- Interactive design should be intuitive: need testing and feedback from others (Project 2 design critique!)
- Seven categories of interactivity

We will learn how to use plotly and ipywidgets to do a majority of these techniques